TryHackMe Room Writeup 3 of 6

TryHackMe: Nmap Basic Port Scans

Network Traffic Analysis, Nmap and Windows Logging | Week 2 | Lab Part

Room detail	Details
Platform	TryHackMe
Room	Nmap Basic Port Scans
Room link	https://tryhackme.com/room/nmap02
Public profile	https://tryhackme.com/p/hariniedward
Status	Completed, 100%
About the room	Learn in depth how the Nmap TCP connect scan, TCP SYN port scan and UDP port scan work.

Objective

Once I know which hosts are alive, the next question is which ports are open on them. This room explains how the TCP connect, SYN and UDP scans work, how to limit and speed up a scan, and how to read the result correctly.

What I learned

- Nmap reports six port states: open, closed, filtered, unfiltered, open|filtered and closed|filtered.
- The TCP flags and the three way handshake decide how each TCP scan reads a port.
- The TCP connect scan (-sT) finishes the whole handshake. It works for any user, but services can log the connection.
- The SYN scan (-sS) sends RST after the SYN-ACK. It is faster and quieter, and needs permission to send raw packets.
- The UDP scan (-sU) has no handshake. A closed port returns an ICMP port unreachable message, and an open port usually says nothing.
- Port selection: -p for one port, a list or a range, -p- for all ports, -F for the most common 100 ports.
- Speed and noise: timing templates from -T0 to -T5, --min-rate, --max-rate and --min-parallelism, with the trade off between speed, noise and accuracy.
- The options -v and -vv show more detail while a scan runs.

Port states explained

State	What it means
open	A service is accepting connections on the port.
closed	The port can be reached, but nothing is listening.
filtered	A firewall or filter is blocking the probe, so Nmap cannot decide.
unfiltered	The port can be reached, but Nmap cannot tell if it is open or closed.
open|filtered	No reply came back, so the port is either open or filtered.
closed|filtered	Nmap cannot tell whether the port is closed or filtered.

The practical task

- Ran connect, SYN and UDP scans against the lab target and compared the results.
- Tried different port options, from a single port to the full range, and saw how the scan time grows.

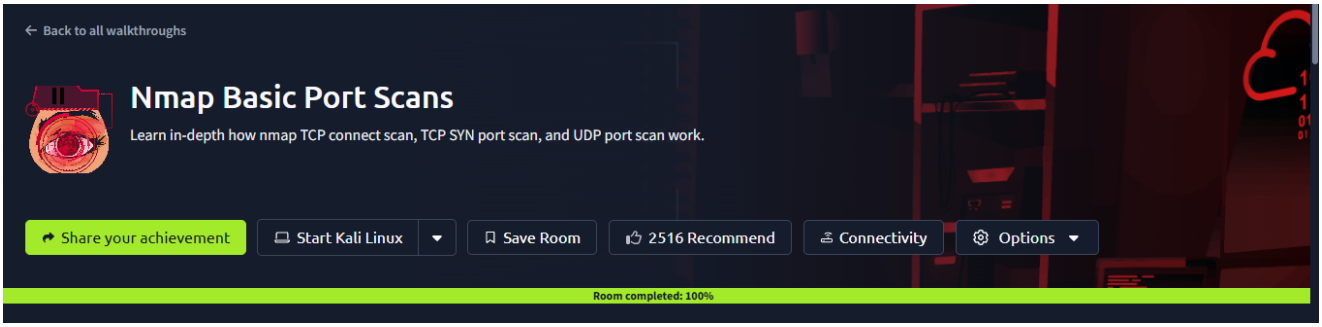
- Changed timing and rate settings to understand the balance between speed, noise and reliability.
- Used the room questions to check that each result matched what the handshake theory predicts.

Why this matters for a SOC analyst

- The same port states describe what an attacker learns, so a defender can judge how much a scan revealed.
- A scan of the full port range, or a very high rate of connection attempts, stands out clearly in firewall and IDS logs.
- A burst of ICMP port unreachable messages can be the footprint of a UDP scan.
- SYN scans leave fewer application logs, which is why network level monitoring matters.

> **Key takeaway**
> Reading port states correctly is the base of both scanning and detection. A closed port answers, a filtered port stays silent, and every scan type draws that line a little differently.

Completion proof



Room page showing 100% completion. My profile details have been removed from the screenshot.

References

Room: https://tryhackme.com/room/nmap02

My public profile: https://tryhackme.com/p/hariniedward